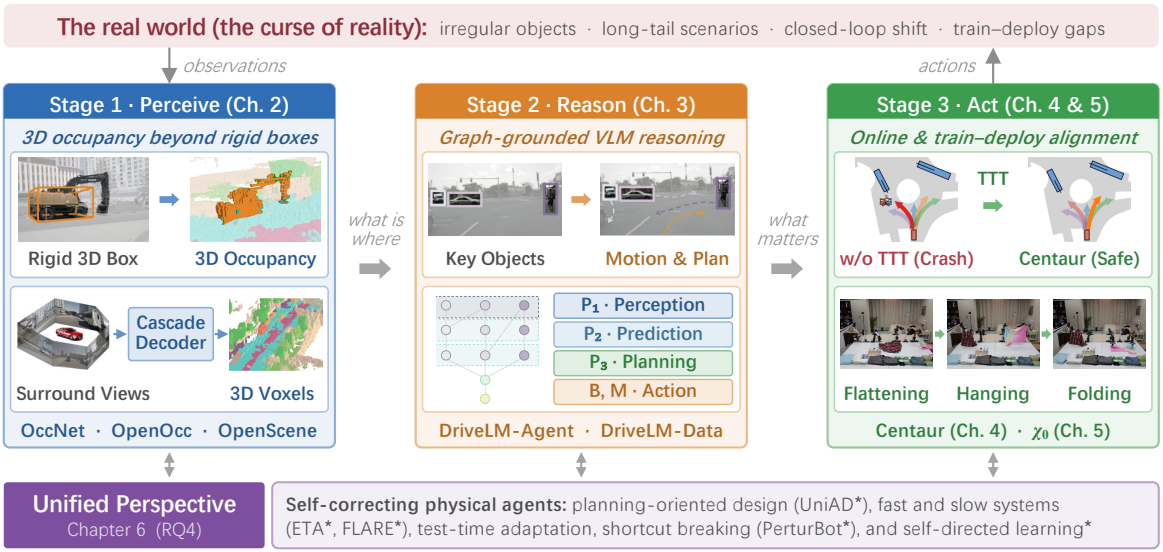

The real world (the curse of reality): irregular objects · long-tail scenarios · closed-loop shift · train–deploy gaps
observations
actions
Stage 1 · Perceive (Ch. 2)
3D occupancy beyond rigid boxes
Rigid 3D Box
3D Occupancy
Surround Views
Cascade Decoder
3D Voxels
OccNet · OpenOcc · OpenScene
what is where
Stage 2 · Reason (Ch. 3)
Graph-grounded VLM reasoning
Key Objects
Motion & Plan
P1 · Perception
P2 · Prediction
P3 · Planning
B, M · Action
DriveLM-Agent · DriveLM-Data
what matters
Stage 3 · Act (Ch. 4 & 5)
Online & train–deploy alignment
TTT
w/o TTT (Crash)
Centaur (Safe)
Flattening
Hanging
Folding
Centaur (Ch. 4) · χ0 (Ch. 5)
Unified Perspective
Chapter 6 (RQ4)
Self-correcting physical agents: planning-oriented design (UniAD*), fast and slow systems (ETA*, FLARE*), test-time adaptation, shortcut breaking (PerturBot*), and self-directed learning*